Dynamical Localization for Random Quantum Walks

Group de Travail

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10/02/2025









Table of Contents

1 Scattering Quantum Walks

2 Dynamical Localization

3 Fractional Moment Criterion

4 Eigenfunction Correlator

Definition of Scattering Quantum Walks

Intuition: Quantum particle moves in *discrete time* on edges of directed graph $G = (V, D)$

$x, y \in V$ adjacent	$x \sim y$
directed edge $y \to x$	$(xy) \in D$
vector in ONB corresponding to (xy)	$\|xy\rangle \in l^2(D)$

Convention: $(yx) \in D \iff (xy) \in D$

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$$l^2(D) = \left\{ \psi = \sum_{x \in V} \sum_{y \sim x} \psi_{xy} \, |xy\rangle \quad \text{s.t.} \quad \sum_{x \in V} \sum_{y \sim x} |\psi_{xy}|^2 < \infty \right\}$$

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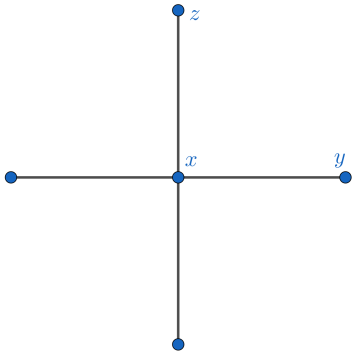
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Define incoming/outgoing subspaces at x:

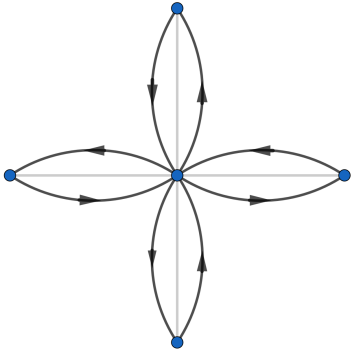
$$\mathcal{H}_x^I = \text{span}\{ \, |xy\rangle \;\; \text{s.t.} \;\; y \sim x \} \;\; \text{and} \;\; \mathcal{H}_x^O = \text{span}\{ \, |yx\rangle \;\; \text{s.t.} \;\; y \sim x \}$$

Incoming and outgoing subspaces



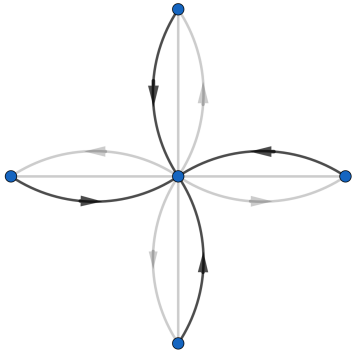
The **undirected** graph at site x

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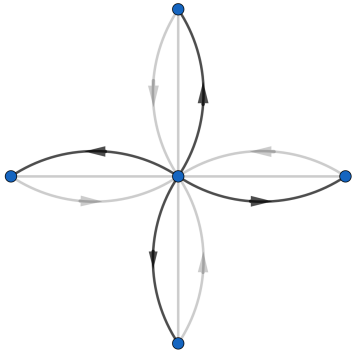
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The subspace $\mathcal{H}_x^I$ of **incoming** edges at x

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The subspace $\mathcal{H}_x^O$ of **outgoing** edges at x

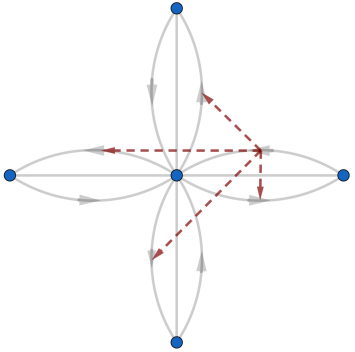
Equip every $x \in V$ with unitary scattering matrix $S(x) : \mathcal{H}_x^I \to \mathcal{H}_x^O$. Define SQW U by its action on incoming edge:

$$U |xy\rangle = \sum_{z \sim x} S_{zy}(x) |zx\rangle$$

This means

$$U = \sum_{x \in V} \sum_{z,y \sim x} S_{zy}(x) |zx\rangle\langle xy|$$

$\Longrightarrow U$ unitary on $l^2(D)$



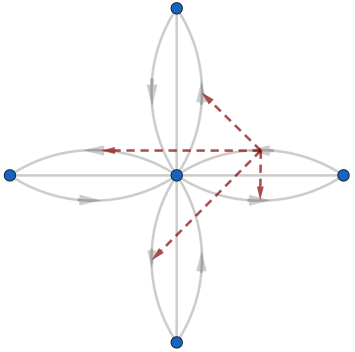
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Given two families of scattering matrices $S(x)$ and $S'(x)$:

$$\|U_{\mathcal{S}} - U_{\mathcal{S}'}\| = \sup_{x\in V} \|S(x) - S'(x)\|_{HS}.$$

Table of Contents

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Full Localization

The choice $S(x) = I$ yields full localization:

$$U\,|xy\rangle = |yx\rangle \quad \forall\,(xy) \in D$$

The SQW U has invariant subspaces

$$\mathcal{H}^{xy} = \mathrm{span}\big(\,|xy\rangle, |yx\rangle\,\big).$$

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Dynamical Localization $\Longrightarrow$ Spectral Localization (i.e. pure point spectrum)
However, if $S(x) \equiv S \neq I$: $\sigma_{cont}(U_S) \neq \varnothing$.

Random Scattering Quantum Walks

Equip every $y \in V$ with RV ω_y st $(\omega_y)_{y \in V}$ is i.i.d. with bounded density $\tau \in L^\infty(\mathbb{T})$. Define

$$D_\omega |xy\rangle = e^{i\omega_y} |xy\rangle \quad \text{and} \quad U_\omega = D_\omega U$$

Equivalent to random scattering matrix $S_\omega(x) = e^{i\omega_x} S(x)$

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Equivalent to random scattering matrix $S_\omega(x) = e^{i\omega_x} S(x)$
If $G = (V, D)$ has bounded degree, we can show

Theorem

For any $g > 0$ there exist $\varphi, c > 0$ such that for any family of scattering matrices with $\|\mathcal{S} - \mathcal{I}\| \leqslant \varphi$, we have for all $|xy\rangle, |x'y'\rangle \in \mathcal{H}$ and $I \subset S^1$

$$\mathbb{E}\Big(\sup_{n\in\mathbb{Z}} |\langle xy \left| U^n P_I(U) \right| x'y'\rangle| \Big) \leqslant c\, e^{-g\, dist(x,x')}. \tag{1}$$

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Fractional Moment Criterion

Idea: Can express $f(U)$ using the resolvent $(U-z)^{-1}$

Theorem (Hamza, Joye, Stolz '09)

If there exist $s \in (0,1)$, $C < \infty$, $g > 0$, $\varepsilon > 0$ *and* $I \subset \mathbb{S}^1$ *such that*

$$\mathbb{E}\Big(\big|\langle xy \,|\, (U_\omega - z)^{-1}\, x'y' \rangle\big|^s\Big) \leqslant C\, e^{-g\, dist(x,x')}$$

for all $|xy\rangle, |x'y'\rangle \in l^2(D)$ *and* $1-\varepsilon < |z| < 1$ *with* $\arg(z) \in I$, *then we have Dynamical Localization as in* (1).

Note: This requires randomness D_ω to be diagonal with i.i.d. phases when expressed in the ONB $\big(|xy\rangle\big)_{x \in V, y \sim x}$

Fractional Moment Criterion

$$D_\omega = \begin{pmatrix} \ddots & & & \\ & \boxed{\begin{matrix} e^{i\omega_x} & & \\ & \ddots & \\ & & e^{i\omega_x} \end{matrix}} & & \\ & & \boxed{\begin{matrix} e^{i\omega_y} & & \\ & \ddots & \\ & & e^{i\omega_y} \end{matrix}} & \\ & & & \ddots \end{pmatrix}$$

$\Longrightarrow$ FMC yields Dyn. Loc. only if the randomness is such that $D_\omega \left|xy\right\rangle = e^{i\omega_{xy}} \left|xy\right\rangle$

FMC for Random SQW

We can show

Theorem

If G has bounded degree, then for any $s \in (0, \frac{1}{3})$ and any $g > 0$, there exist $\varphi, c > 0$ such that for any collection of scattering matrices $(S(x))_{x \in V}$ with $\|\mathcal{S} - \mathcal{I}\| \leqslant \varphi$ we have

$$\sup_{|z| \in [1/2, 3/2] \setminus \{1\}} \mathbb{E}\left(|\langle xy | (U - z)^{-1} | x'y' \rangle|^s \right) \leqslant c e^{-g\, dist(x, x')}, \tag{2}$$

for any two basis vectors $|xy\rangle, |x'y'\rangle \in l^2(D)$.

Sketch of proof of FMC

Key ingredients:

1. Fractional Moment Bound:

$$\sup_{|z|\neq 1} \int_{\mathbb{T}} \int_{\mathbb{T}} |\langle xy \,|\, (U_\omega - z)^{-1} x'y' \rangle|^s \, d\mu(\omega_y) \, d\mu(\omega_{y'}) \leqslant C(s)$$

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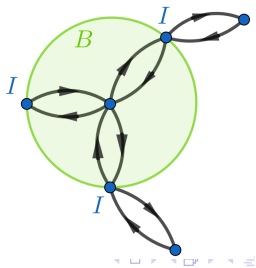
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2. Can restrict U to ball $B = \{v \in V \mid \mathrm{dist}(v, y') \leqslant r\}$ using reflective boundary conditions:

$$\tilde{S}(v) = \begin{cases} I & \text{if } \mathrm{dist}(v, y') = n \\ S(v) & \text{else} \end{cases}$$

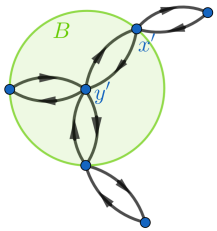
Let U^B and U^{B^c} be the restrictions to B and B^c, $T^B = U - U^B \oplus U^{B^c}$

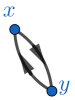


Sketch of proof of FMC

Let B be circle of radius 1 centered at y'
Resolvent identity $R = R^{B\oplus B^c} - R^{B\oplus B^c} T^B R$

$$|\langle xy\,|R\,x'y'\rangle|^s = |\langle xy\,|R^{B\oplus B^c} T^B R\,x'y'\rangle|^s$$
$$\leqslant \sum_{v\in\partial B, u, v'\sim v} |\langle xy\,|R^{B^c}\,uv\rangle|^s\,|\langle uv\,|T^B vv'\rangle|^s\,|\langle vv'\,|R\,x'y'\rangle|^s$$

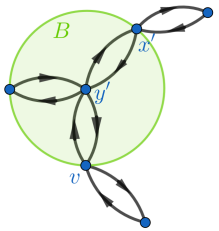




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Bound T^B by $\|I - \mathcal{S}\| = \sup_{x\in V} \|I - S(x)\|_{HS}$
Integrating over randomness in B bounds R term:

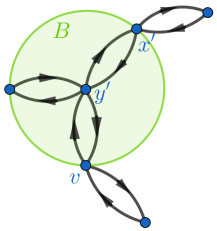
$$\mathbb{E}(|\langle xy\,|R\,x'y'\rangle|^s) \leqslant \sum_{v\in\partial B, u\sim v} d\,C_s \|I-\mathcal{S}\|^s\,\mathbb{E}(|\langle xy\,|R^{B^c}\,uv\rangle|^s)$$



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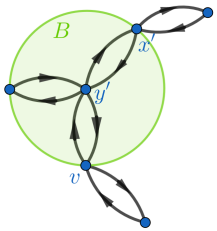
Set $r(v) = \max_{u\sim v} \mathbb{E}(|\langle xy\,|R\,uv\rangle|^s)$. If $q := d^3 C_s \|I-\mathcal{S}\|^s < 1$, then

$$r(y') \leqslant q r(v) \leqslant \ldots \leqslant C\,q^{\mathrm{dist}(y,y')}$$

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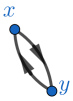
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Randomness on boundary ∂B affects both B and B^c
Need to decouple randomness in- and outside of ball

Sketch of proof of FMC

Randomness on boundary ∂B affects both B and B^c
Need to decouple randomness in- and outside of ball
$\Longrightarrow$ resampling arguments $\square$

However: We can not immediately conclude Dynamical Localization from FMC, as we do not have "enough randomness"

What's the problem?

Assume $D_\omega |xy\rangle = e^{i\omega_{xy}} |xy\rangle$

At one point in proof of "FMC $\Longrightarrow$ Dyn Loc" we restrict to randomness at (xy) :

$$P = |xy\rangle\langle xy|$$

We have two operators A and B that are related by

$$A = \big(I + AP\big)B$$

When computing the $\langle xy| \cdot |x'y'\rangle$ matrix element, we can solve the equation

$$\langle xy|B\,x'y'\rangle = \frac{\langle xy|A\,x'y'\rangle}{1 + \langle xy|\,A\,xy\rangle}$$

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However, if $D_\omega |xy\rangle = e^{i\omega_y} |xy\rangle$, then

$$P = \sum_{v\sim y} |vy\rangle\langle vy|$$

and we can not solve for $\langle xy|B\,x'y'\rangle$.

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Eigenfunction Correlator

The Eigenfunction correlator Q at $(xy), (x'y') \in D$ on the interval $I \subset \mathbb{S}^1$ is

$$Q(xy, x'y'; I) = \sup_{\substack{F \in C_c(I) \\ \|F\|_\infty \leqslant 1}} \left| \langle xy | F(U) x'y' \rangle \right|.$$

Taking $F(z) = z^n$ yields bound for Dynamical Localization
If $U = \sum_\alpha \lambda_\alpha P_\alpha$ has discrete spectral decomposition, then

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Wlog finite graph: If $\lim_{L\to\infty} \| T^{B_L} |xy\rangle \| = 0$ for all $(xy) \in D$, then

$$Q(xy, x'y'; I) \leqslant \liminf_{L\to\infty} Q^{B_L}(xy, x'y'; I)$$

Relation to FMC

Can show under the right assumptions (*Fractional Moment Bound* and *Spectral Averaging*): for a ball B st $(xy) \in B$, $(x'y') \notin B$:

$$\mathbb{E}\left[Q(xy, x'y'; I)\right] \leqslant C \sum_{(uv)\in\partial B} \sup_{\delta\in(0,1)} \left\{ \int_I \mathbb{E}\left[|\langle x'y'|(U^B - \delta e^{i\theta})^{-1} uv\rangle|^s \right] \frac{d\theta}{2\pi} \right\}$$

So if boundary ∂B does not grow "too fast", we obtain exponential decay of Q in $\mathrm{dist}(x, x')$

FMC $\Longrightarrow$ Exponential decay of Eigenfunction correlator $\Longrightarrow$ Dynamical Localization

Thank you for your attention!